Project Scope Management

What is Project Scope Management?

- **Scope** refers to everything that needs to be done in a project- the work, activities, and final results. It clearly defines **what is included** in the project and **what is not included**.

- A **deliverable** is a product produced as part of a project, such as hardware or software, planning documents, or meeting minutes

- Project scope management includes the processes involved in defining and controlling what is or is not included in a project

Project Scope Management Processes

- **Planning scope management**: Create a scope management plan that explains how the scope will be defined, validated, and controlled.
- **Collecting requirements**: defining and documenting the features and functions of the products produced during the project, as well as the processes used for creating them
- **Defining scope**: reviewing the project charter, requirements documents, and organizational process assets to create a scope statement
- **Creating the WBS**: subdividing the major project deliverables into smaller, more manageable components
- **Validating scope**: formalizing acceptance of the project deliverables
- **Controlling scope**: controlling changes to project scope throughout the life of the project

Project Scope Management Summary

Planning
Process: **Plan scope management**
Outputs: Scope management plan, requirements management plan
Process: **Collect requirements**
Outputs: Requirements documentation, requirements traceability matrix
Process: **Define scope**
Outputs: Project scope statement, project documents updates
Process: **Create WBS**
Outputs: Scope baseline, project documents updates

Monitoring and Controlling
Process: **Validate scope**
Outputs: Accepted deliverables, change requests, work performance information, project documents updates
Process: **Control scope**
Outputs: Work performance information, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start **Project Finish**

Planning Scope Management

- The project team uses expert judgment and meetings to develop two important outputs: the scope management plan and the requirements management plan
- The scope management plan is a subsidiary part of the project management plan

Scope Management Plan Contents

- How to prepare a detailed project scope statement
- How to create a WBS
- How to maintain and approve the WBS
- How to obtain formal acceptance of the completed project deliverables
- How to control requests for changes to the project scope

Requirements Management Plan

- The PMBOK® Guide, Fifth Edition, describes requirements as “conditions or capabilities that must be met by the project or present in the product, service, or result to satisfy an agreement or other formally imposed specification”

- The **requirements management plan** documents how project requirements will be analyzed, documented, and managed

Collecting Requirements

- For some IT projects, it is helpful to divide requirements development into categories called:
  - Elicitation (ask and collect information),
  - Analysis (understand and organize what was collected),
  - Specification (documenting the requirements properly), and
  - Validation (Check and confirm with stakeholders.)
- It is important to use an iterative approach to defining requirements since they are often unclear early in a project

Methods for Collecting Requirements

- Interviewing
- Focus groups and facilitated workshops
- Questionnaires and surveys
- Observation
- Prototyping
- **Brainstorming**
- **Benchmarking**, or generating ideas by comparing specific project practices or product characteristics to those of other projects or products inside or outside the performing organization, can also be used to collect requirements

Requirements Traceability Matrix

- A **requirements traceability matrix (RTM)** is a table that lists requirements, various attributes of each requirement, and the status of the requirements to ensure that all requirements are addressed
- Table 5-1. Sample entry in an RTM

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet requirement by having 4GB of memory.

Defining Scope

- **Project scope statements:** A formal document that describes the project's scope, deliverables, boundaries, assumptions, and constraints. It provides a common agreement among stakeholders about the project's outcomes.
- As time progresses, the scope of a project should become clearer and specific

Sample Project Charter (partial)

Project Title: Information Technology (IT) Upgrade project

Project Start Date: March 4
Project Finish Date: December 4

Key Schedule Milestones:
- Inventory update completed April 15
- Hardware and software acquired August 1
- Installation complete October 1
- Testing complete November 15

Budget information: Budgeted $1,000,000 for hardware and software costs and $500,000 for labor costs.

Manager: Kim Nguyen, (310) 555- 2784, knguyen@course.com

Project objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers as well as associated network hardware and software.

Main Project Success Criteria: The hardware, software, and network upgrades must meet all written specifications, be thoroughly tested, and be completed in nine months. Employee work disruptions will be minimal.

Approach:
- Update the IT inventory database to determine upgrade needs.
- Develop detailed cost estimate for project and report to CIO
- Issue a request for quote to obtain hardware and software
- Use internal staff as much as possible for planning, analysis, and installation

Adding Project Scope

Project Charter: Upgrades may affect servers…(listed under Project Objective)
Project Scope Statement, Version 1: Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least 2 weeks before installation.
Project Scope Statement, Version 2 Servers: This project will required purchasing ten new servers to support Web, network, database, applications and printing functions. Virtualization will be used to maximize efficient. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.

Creating the Work Breakdown Structure (WBS)

- A **Work Breakdown Structure (WBS)** is a hierarchical breakdown of the total project work into **smaller, manageable pieces**.

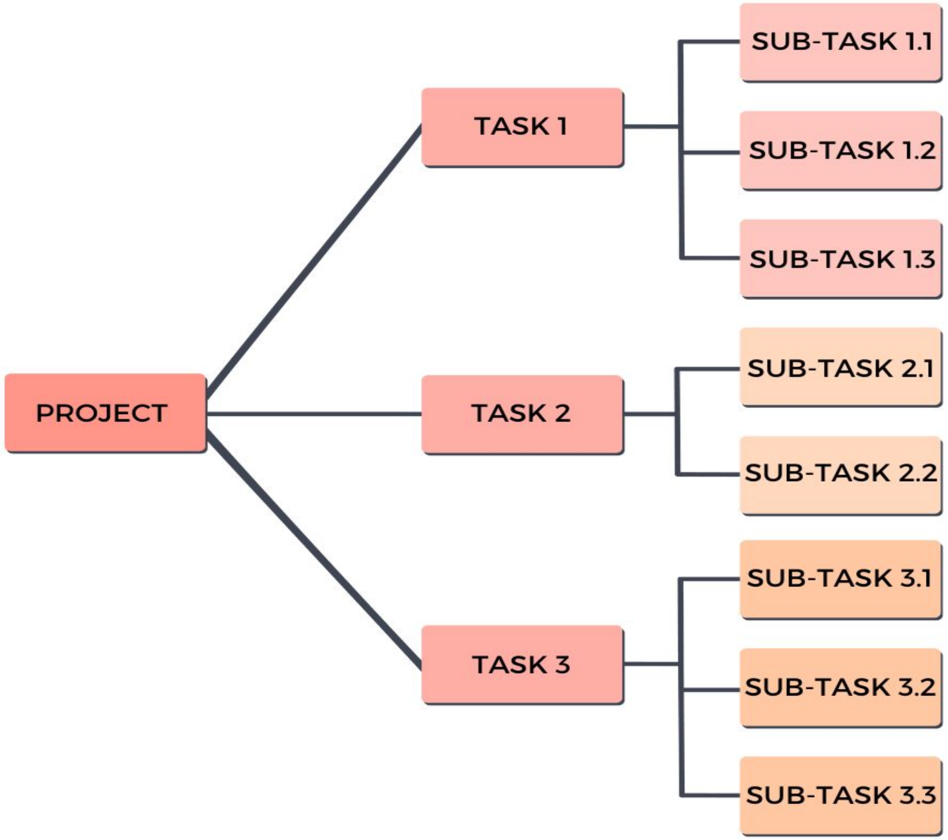
- **Decomposition** is subdividing project deliverables into smaller pieces
- A **work package** is a task at the lowest level of the WBS

- In simple words, **WBS = Breaking the big project into smaller tasks until they are easy to manage.**

Structure of WBS

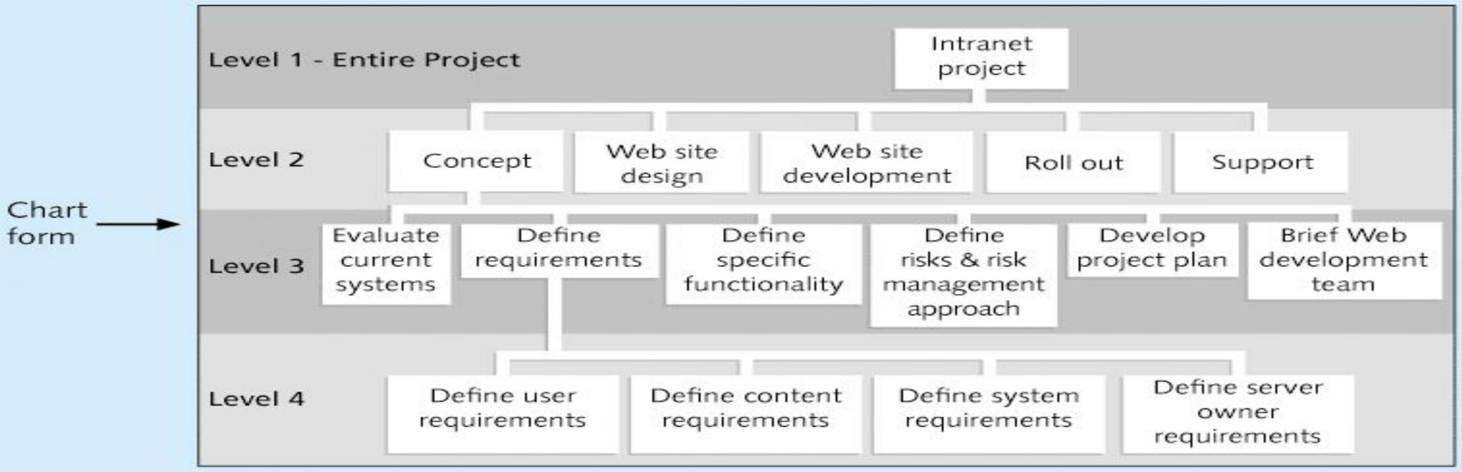
- **Level 1:** The overall project.
- **Level 2:** Major deliverables or phases.
- **Level 3:** Smaller components, sub-deliverables, or work packages.
- **Level 4 (and beyond):** Detailed tasks if needed.

Work Breakdown Structure Example




```
4.0 Development
    ├── 4.1 Frontend Development
    │       ├── 4.1.1 Homepage
    │       ├── 4.1.2 Product Pages
    │       └── 4.1.3 Shopping Cart & Checkout
    ├── 4.2 Backend Development
    │       ├── 4.2.1 User Authentication
    │       ├── 4.2.2 Product & Inventory Management
    │       └── 4.2.3 Order Processing System
    └── 4.3 Payment Gateway Integration
```

Sample Intranet WBS Organized by Phase



Tabular form with PMI numbering

1.1 Concept
- 1.1.1 Evaluate current systems
- 1.1.2 Define requirements
  - 1.1.2.1 Define user requirements
  - 1.1.2.2 Define content requirements
  - 1.1.2.3 Define system requirements
  - 1.1.2.4 Define server owner requirements
- 1.1.3 Define specific functionality
- 1.1.4 Define risks and risk management approach
- 1.1.5 Develop project plan
- 1.1.6 Brief Web development team

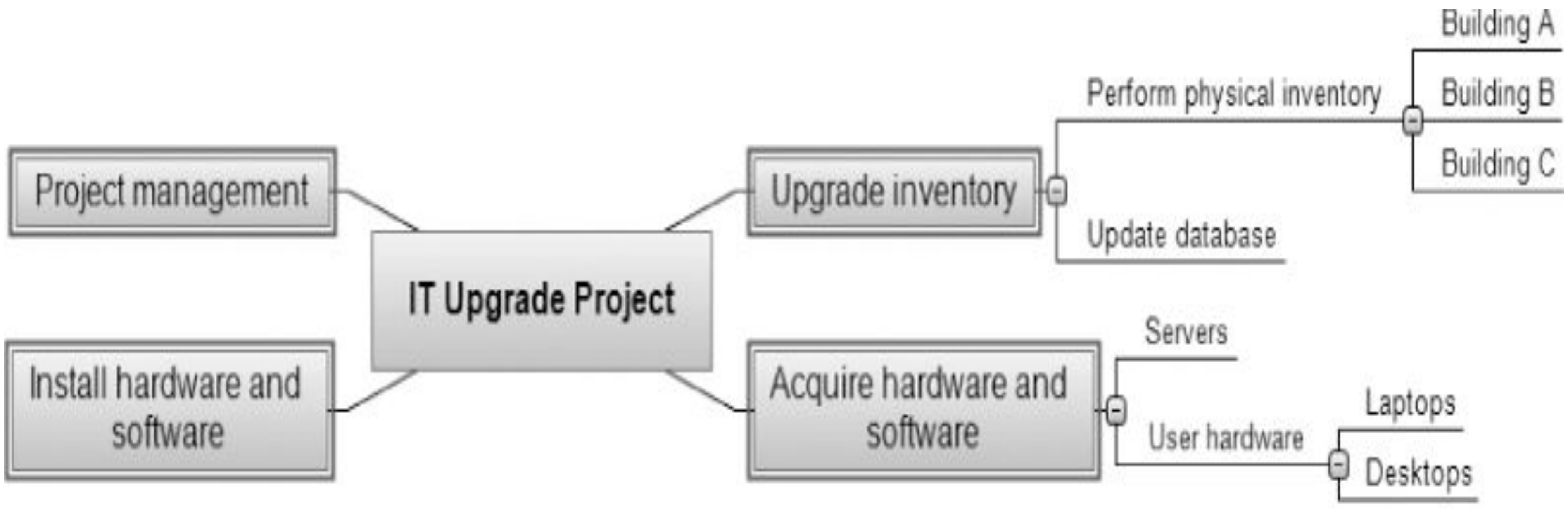
1.2 Web site design
1.3 Web site development
1.4 Roll out
1.5 Support

Approaches to Developing WBSs

- **Using guidelines:** Some organizations, like the PMI (Project Management Institute), provide guidelines for preparing WBSs

- The **analogy approach**: Review WBSs of similar projects and tailor to your project
- The **top-down approach**: Start with the largest items of the project and break them down
- The **bottom-up approach**: Start with the specific tasks and roll them up
- Mind-mapping approach: **Mind mapping** is a technique that uses branches radiating out from a core idea to structure thoughts and ideas

Sample Mind-Mapping Approach for Creating a WBS



Source: MatchWare's MindView 4 Business Edition

The WBS Dictionary and Scope Baseline

- Many WBS tasks are vague and must be explained more so that people know what to do and can estimate how long it will take and what it will cost to do the work
- A **WBS dictionary** is a document that describes detailed information about each WBS item

Sample WBS Dictionary Entry

WBS Dictionary Entry March 20
Project Title: Information technology (IT) Upgrade Project
WBS Item Number: 2.2
WBS Item Name: Update Database
Description: The IT department maintains an online database of hardware and software on the corporate intranet. However, we need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Perform Physical Inventory, and must precede WBS Item Number 3.0, Acquire Hardware and Software.

Validating Scope

- **Validate Scope** is the process of **formalizing acceptance** of the completed project deliverables.
- Acceptance is often achieved by a customer inspection and then sign-off on key deliverables

- Validating scope means checking deliverables with stakeholders, getting formal approval, and making sure the project is building exactly what was promised.

Controlling Scope

- Scope control involves controlling changes to the project scope
- Control Scope means keeping the project on track by only working on approved requirements, and carefully managing any requested changes.

Example (IT Project – Mobile App)

- Scope baseline: App must have login, profile, and messaging features.
- During execution, a stakeholder asks for "dark mode" (not in requirements).
- **Control Scope** process:
  - Evaluate request.
  - Check impact on time, cost, resources.
  - Approve or reject formally.
  - Update documents if approved.

Summary

- Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
- Main processes include
  - Define scope management
  - Collect requirements
  - Define scope
  - Create WBS
  - Validate scope
  - Control scope